Enthusiasts often look for PSUs with high compliance ratings, as these indicators of quality contribute to better performance and energy efficiency.

Connector Types
Connector types refer to the various power connectors provided by a PSU, such as 24-pin ATX, 4/8-pin CPU, PCIe, SATA, and Molex. The number and type of connectors affect compatibility with components. Enthusiasts evaluate connector types to ensure they have enough connections for all their hardware, avoiding bottlenecks and ensuring stable power delivery to each component in their system.

Continuous Power Output
Continuous power output is the amount of power a PSU can deliver consistently without overheating or shutting down. This specification is crucial for ensuring stable operation under heavy loads. Enthusiasts examine continuous power ratings to ensure that their PSU can support peak performance during intensive tasks, like gaming or video editing, without risk of failure.

Efficiency Rating
Efficiency rating indicates how effectively a PSU converts AC power from the wall into usable DC power for the computer. This rating is often classified by standards like 80 PLUS, which categorizes efficiency levels (e.g., Bronze, Silver, Gold, Platinum). Higher efficiency ratings result in lower energy waste, cooler operation, and reduced electricity bills. Enthusiasts prioritize high-efficiency PSUs to enhance system performance and sustainability.

Form Factor
Form factor refers to the physical size and shape of the PSU, which must match the case and motherboard specifications. Common form factors include ATX, SFX, and TFX. Enthusiasts choose PSUs with appropriate form factors to ensure compatibility with their build, particularly in compact cases where space is limited, as mismatched components can lead to installation difficulties.

Hold-Up Time
Hold-up time refers to the duration a PSU can maintain its output voltage after losing AC input power, typically measured in milliseconds (ms). A longer hold-up time allows the system to remain powered during brief interruptions, preventing data loss or corruption. Enthusiasts consider hold-up time when selecting PSUs for systems that may be subject to unstable power sources, ensuring reliability during power fluctuations.

Inrush Current
Inrush current is the initial surge of current drawn by a PSU when it is first powered on. This spike can be several times higher than the PSU's rated output. Enthusiasts consider inrush current when selecting PSUs for systems with multiple high-power components, as excessive inrush current can trip circuit breakers or damage electrical components. Advanced PSUs feature soft-start technology to mitigate this effect.

Lifetime Rating
Lifetime rating indicates the expected lifespan of a PSU, typically expressed in hours at a specified temperature (e.g., 100,000 hours at 25°C). This rating helps consumers assess the reliability and durability of a PSU. Enthusiasts often prioritize PSUs with high lifetime ratings to ensure long-term performance and reduce the risk of failure in demanding applications, particularly for systems subjected to heavy workloads.

Modular Design
Modular design refers to the ability to attach or detach power cables from the PSU. There are three types: non-modular, semi-modular, and fully modular. Fully modular PSUs allow for complete customization, reducing cable clutter and improving airflow within the case. Enthusiasts favor modular PSUs for aesthetic reasons and easier cable management, particularly in builds with tempered glass panels where visual appeal is important.

Over-Voltage Protection (OVP)
OVP is a safety feature that protects components from excessive voltage levels that could cause damage. If the voltage exceeds safe thresholds, the PSU automatically shuts down to protect connected hardware. Enthusiasts seek PSUs with effective OVP to safeguard their systems from power surges, ensuring longevity and stability during operation.

PFC (Power Factor Correction)
PFC is a technology that improves the power factor of a PSU, helping to reduce energy loss and increase efficiency. Active PFC is more efficient than passive PFC, resulting in better performance and reduced electricity costs. Enthusiasts prefer PSUs with active PFC to minimize energy waste and improve overall system performance, especially in high-demand scenarios.

MOTHERBOARD

ATX (Advanced Technology eXtended)
The ATX form factor, introduced by Intel in 1995, is the most common motherboard size. It measures 305 mm x 244 mm (12" x 9.6") and offers multiple expansion slots, ports, and connectors. Enthusiasts favor ATX motherboards for their flexibility and expandability, providing ample room for components such as GPUs and additional PCIe cards. The design allows for efficient airflow, making it a popular choice for gaming and high-performance systems.

BIOS/UEFI
BIOS (Basic Input/Output System) and UEFI (Unified Extensible